

Ryan Cho

Senior AI Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full stack engineer with 8+ years building production AI systems, scalable platforms, and secure data pipelines in regulated finance and SaaS. Expert in Python, Node.js, LLM applications, RAG architectures, vector databases, and model evaluation with sensitivity and specificity benchmarking. Led cross-functional delivery, mentored junior engineers, and collaborated with product and executive stakeholders on architectural strategy, rapid AI prototyping, and cloud-native deployment.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and deployed **end-to-end LLM-powered decision tools** with Python, LangChain, and vector databases, enabling natural-language search over 2M+ unstructured documents for portfolio analysts.
- Built **retrieval-augmented generation pipelines** with embedding optimization, prompt caching, and A/B benchmarking that improved answer relevance by 34% while reducing per-query inference cost.
- Led **full stack delivery** of React and Node.js dashboards integrating real-time model inference APIs, supporting 500+ daily active users at 99.9% availability on AWS EKS.
- Designed **secure, compliance-ready data pipelines** for structured and unstructured document ingestion with encryption at rest, RBAC, and audit trails aligned to regulated-environment standards.
- Mentored **three junior engineers** on production ML lifecycle practices, including model evaluation with AUC-ROC metrics, code review standards, and deployment guardrails.
- Partnered with **product and executive stakeholders** to rapidly prototype AI agent workflows, validating 8 concepts through POC cycles before committing engineering investment.
- Implemented **CI/CD automation** with Docker, Kubernetes, and GitHub Actions, reducing deployment time from 45 minutes to under 15 minutes and enabling daily model refresh.
- Developed **NLP document parsing services** using PyTorch for semi-structured report extraction, automating entity recognition across 12 document types and cutting manual review by 60%.
- Established **model evaluation frameworks** with sensitivity, specificity, and regression testing suites that gate promoted models and document performance for audit review.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Delivered **scalable backend services** in Python and Java for trade processing and risk workflows, handling 50K+ daily transactions with sub-200ms p95 latency.
- Built **RESTful APIs and data integration layers** connecting internal platforms to third-party market data providers, reducing manual reconciliation effort across 4 downstream systems.
- Contributed to **cloud migration initiatives** on AWS, containerizing legacy services with Docker and establishing foundational CI/CD patterns adopted by two engineering teams.
- Implemented **data validation and pipeline monitoring** for batch ETL jobs processing structured financial datasets, improving data quality defect detection by 40%.
- Collaborated with **cross-functional analysts and compliance teams** to design software adhering to regulatory controls, access policies, and change-management requirements.
- Optimized **SQL query performance** and indexing strategies across PostgreSQL databases serving reporting dashboards used by 200+ internal stakeholders.
- Supported **production incident resolution** for tier-1 banking applications, participating in on-call rotation and root-cause analysis for critical-path outages.

- Authored **technical design documents and interface specifications** for new trading platform features, guiding implementation across 3 scrum teams.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Developed **React and Node.js features** for a SaaS compliance platform, shipping user-facing workflows that improved checkout audit completion rates.
- Integrated **third-party REST APIs** for payment and receipt data ingestion, building reliable retry logic and error handling for production traffic.
- Wrote **unit and integration tests** with Jest and Mocha, increasing frontend test coverage from 45% to 78% during the internship tenure.
- Participated in **Agile ceremonies** and code reviews, delivering 15+ pull requests merged to main with guidance from senior engineers.
- Refactored **legacy JavaScript modules** into modular ES6 components, reducing page load time and improving maintainability of shared UI patterns.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive web pages** with HTML, CSS, and JavaScript for a nonprofit fundraising site, improving mobile usability across core donation flows.
- Implemented **CMS-driven content updates** using WordPress, enabling non-technical staff to publish event pages without developer support.
- Optimized **front-end performance** through image compression and lazy loading, reducing average page load time on key landing pages.
- Collaborated with **design and outreach stakeholders** to iterate on UI feedback and ship updates within a fast-paced two-month engagement.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML/CSS, Next.js, responsive UI design

Backend: Python, Node.js, FastAPI, Express, REST APIs, microservices, LLM integration

Databases: PostgreSQL, MongoDB, Redis, vector databases (Pinecone, pgvector), Elasticsearch

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Terraform

Monitoring & Tools: LangChain, LlamaIndex, TensorFlow, PyTorch, Hugging Face, OpenAI API, Git, Datadog

Workflow & PM: Agile, Jira, cross-functional leadership, technical mentorship, stakeholder communication